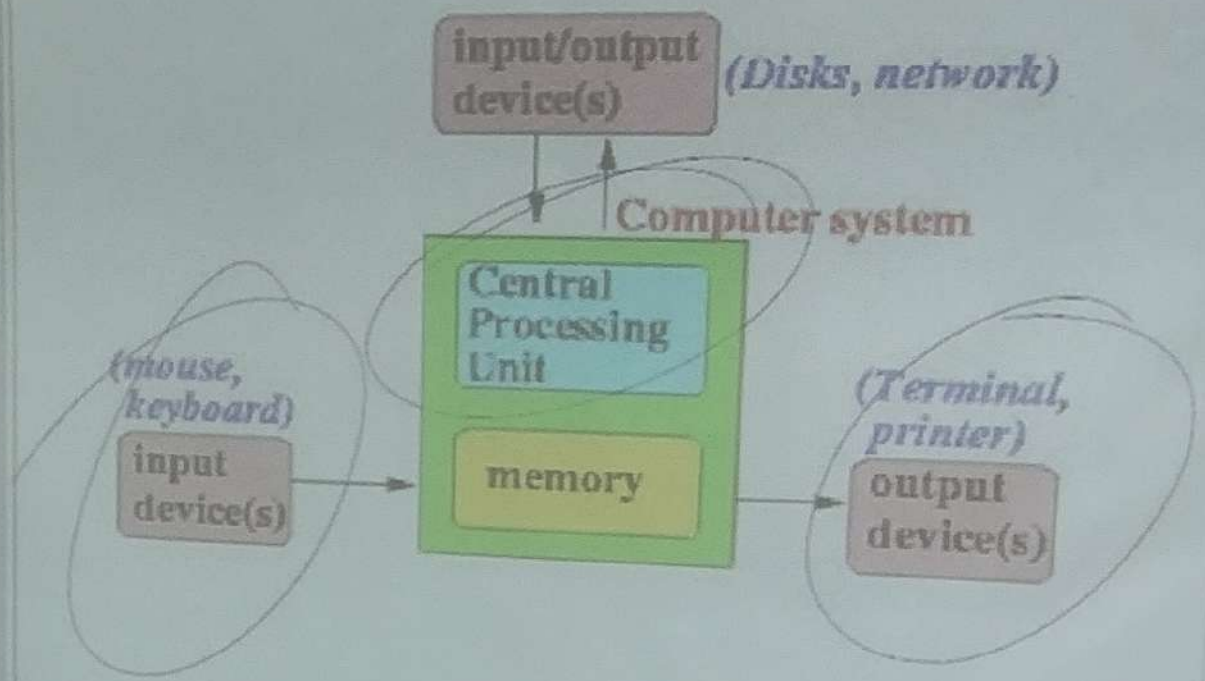
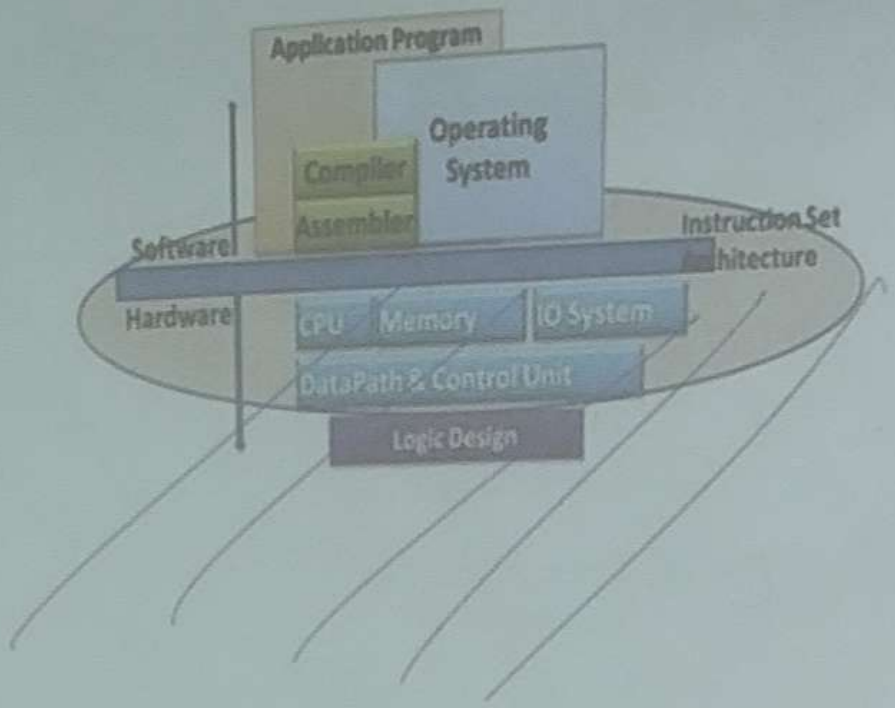
Functional Components of a Computer

- The basic functional components or elements of a digital computer system basically has the hardware and software.
- The hardware is the physical component/part such as a keyboard, mouse, monitor, etc.
- The software is the set of programs and instructions which perform several specific operations.
- *Both hardware and software together act as functional components.*
- They help to complete the functional cycle which consists of input, processing, and output.
- We learn about the different functional components of a digital computer and their working and interconnections.

Functional Components of a Computer



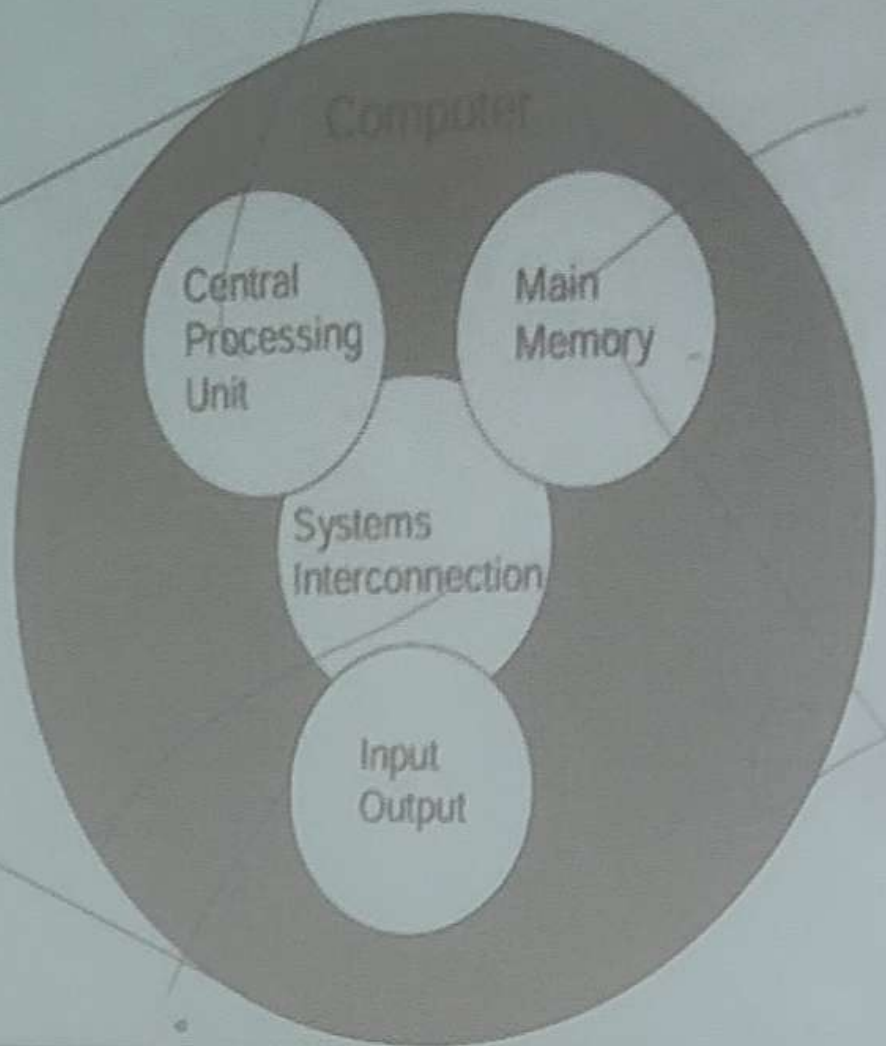
Functional Components of a Computer



Structure

- A computer is a complex system; contemporary computers contain millions of elementary electronic components.
- How, then, can one clearly describe them?
- The key is to recognize the hierarchical nature of most complex systems, including the computer.
- A hierarchical system is a set of interrelated subsystems;
- Each subsystem may, in turn, contain lower level subsystems, until we reach some lowest level of elementary subsystem.
- **Structure** *is the way in which components relate to each other, i.e., are interrelated.*

Structure - Top Level



Functions in Structural View

- The hierarchical nature of complex systems is essential to both their design and their description.
- The designer need only deal with a particular level of the system at a time.
- At each level, the system consists of a set of components and their inter-relationships.
- The behavior at each level depends only on a simplified, abstracted characterization of the system at the next lower level.

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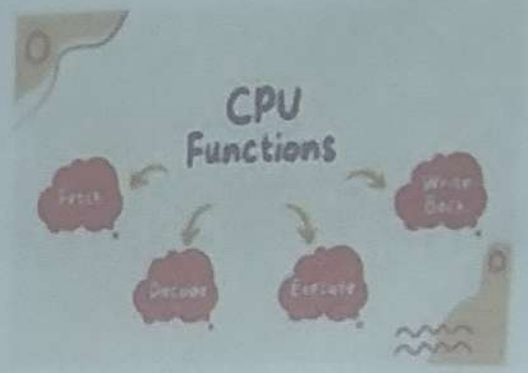
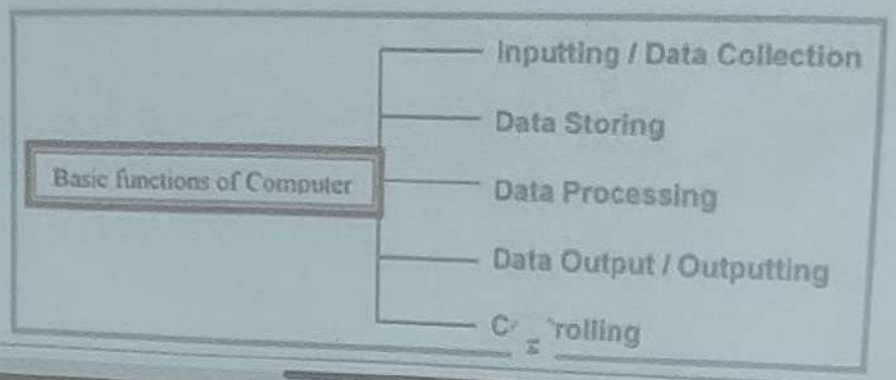
Function

- *Function is the operation of individual components as part of the computer structure.*

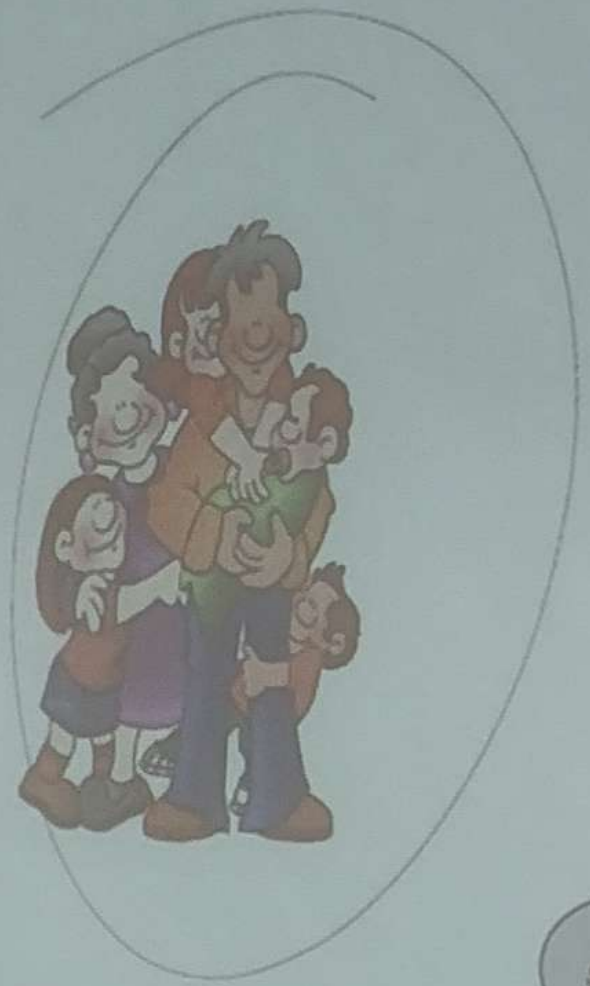
Computer Functional View

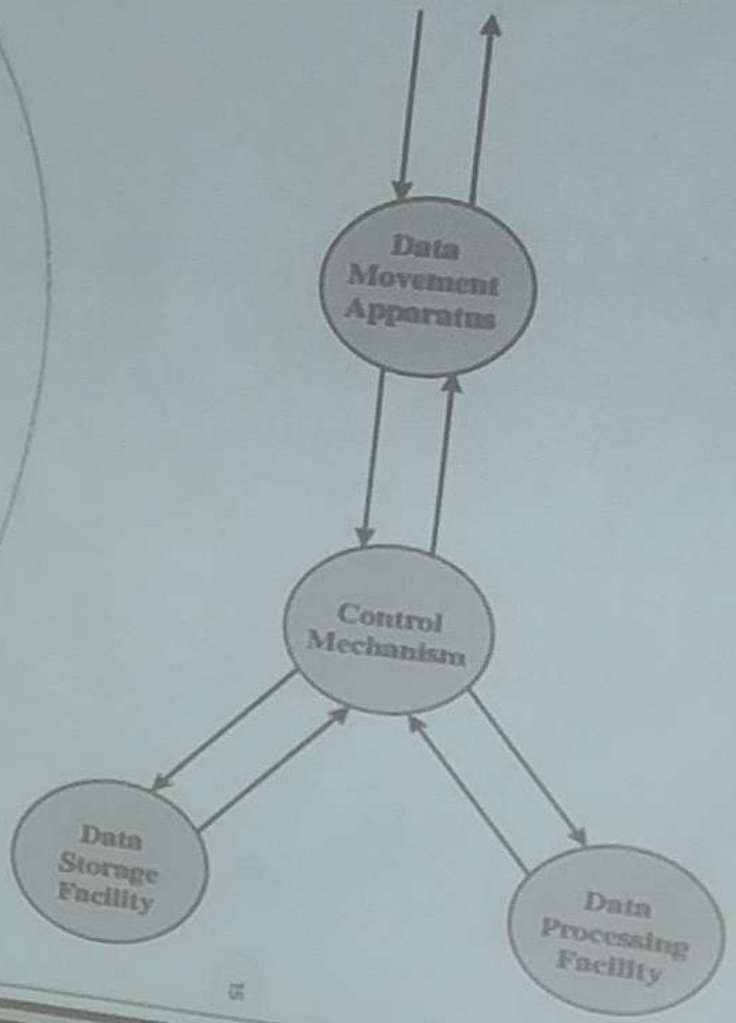
- All computer functions are:
  a) Data movement
  b) Data storage
  c) Data processing
     - From/to storage
     - From/to I/O
  d) Control



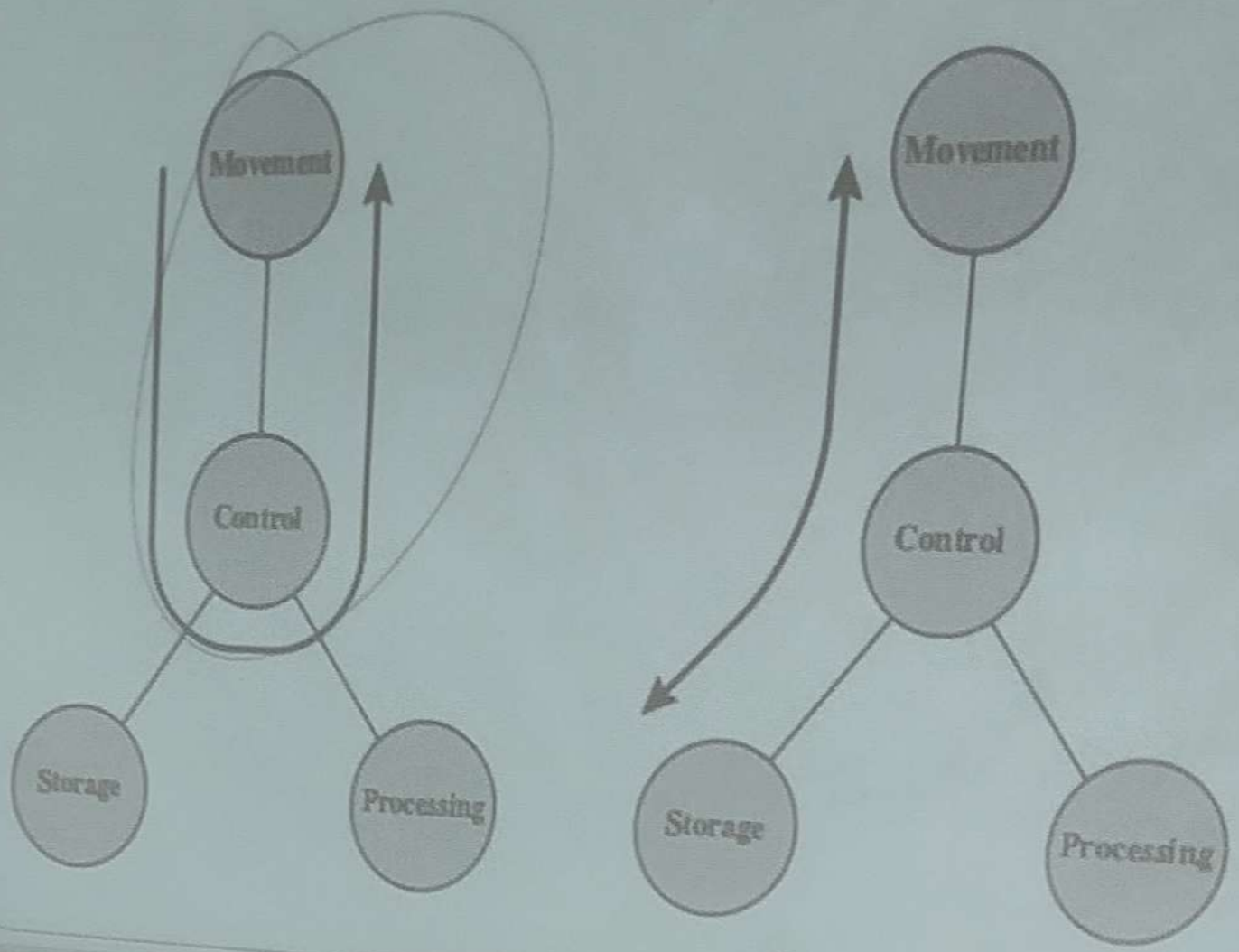


Functional View of a Computer

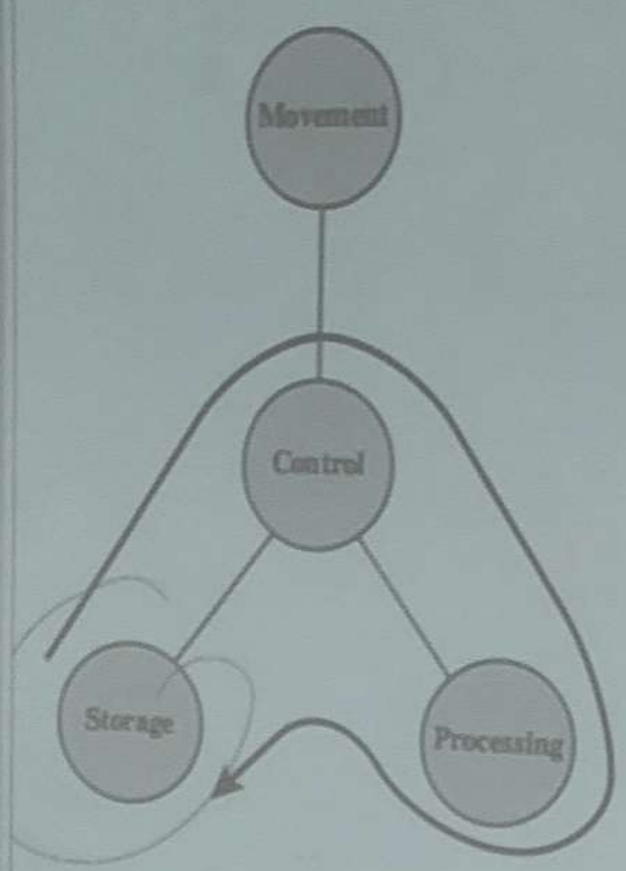




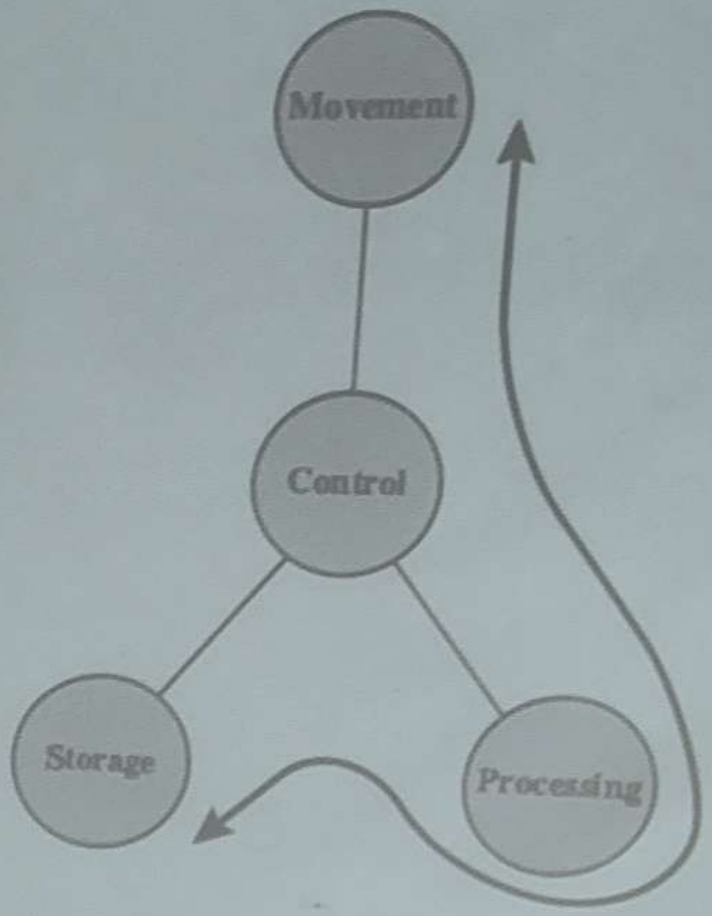
Computer Operations (a) and (b)



Computer Operation (c)



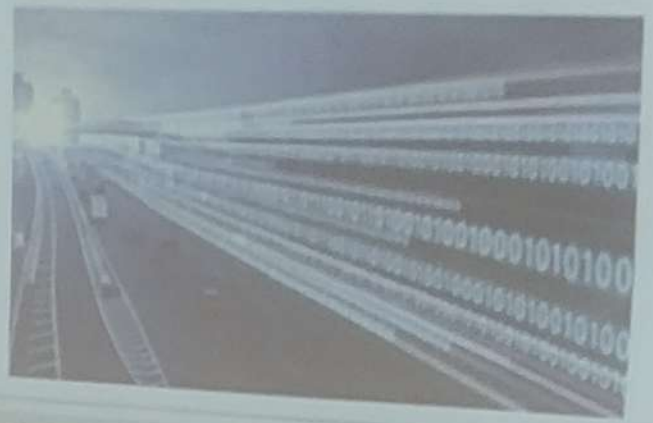
From/To Storage

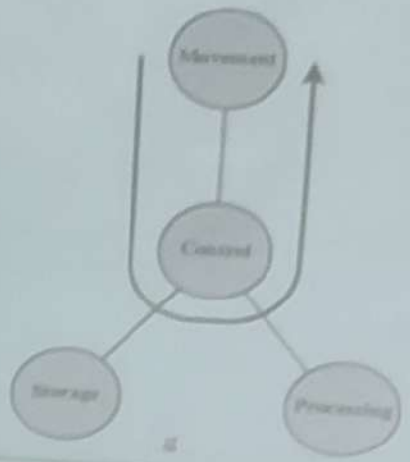


From/To I/O

Operation: Data Movement

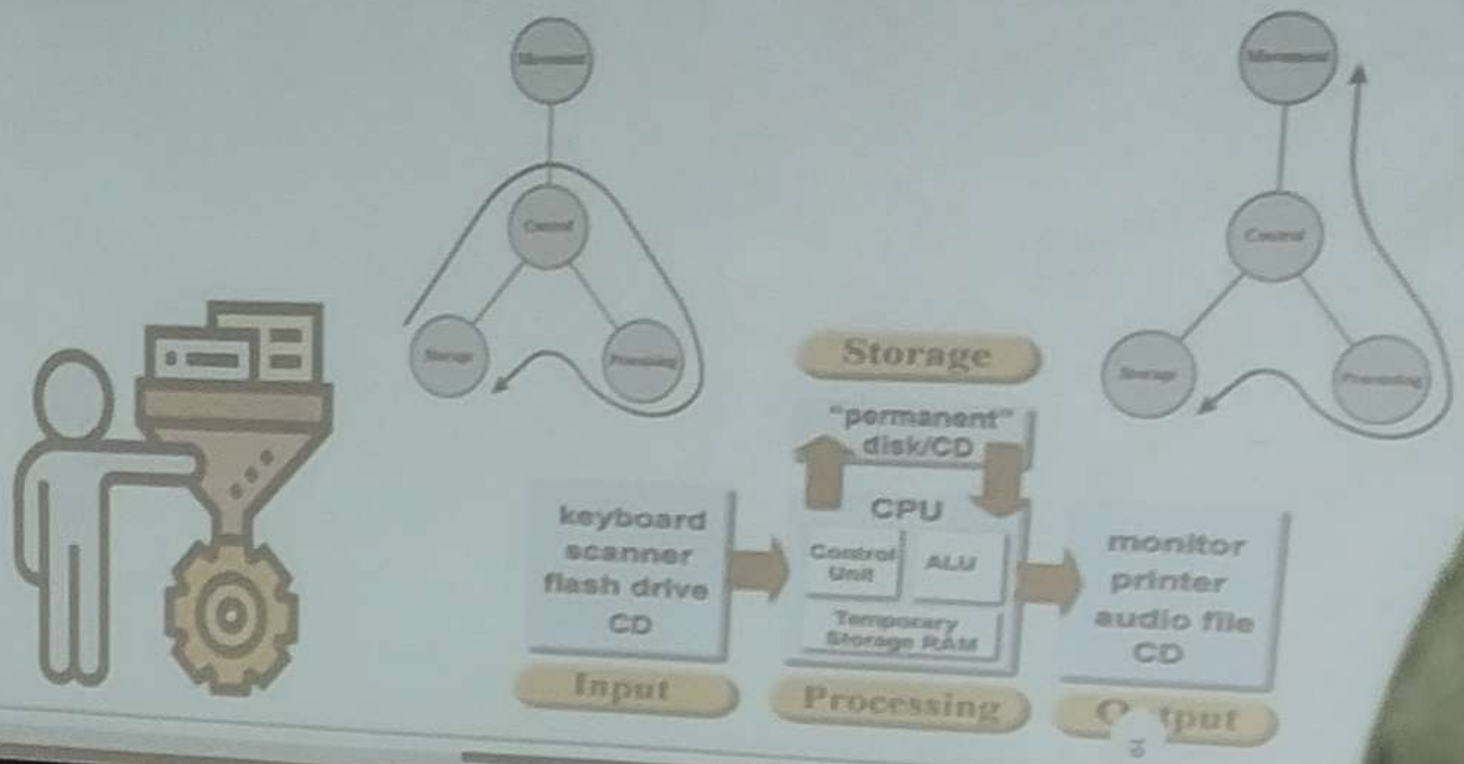
- The computer must be able to move data between itself and the outside world.
- The computer's operating environment consists of devices that serve as either sources or destinations of data.
- When data are received from or delivered to a device that is directly connected to the computer.
- *The process is known as input–output (I/O), and the device is referred to as a peripheral.*
- *When data are moved over longer distances, to or from a remote device, the process is known as data communications.*





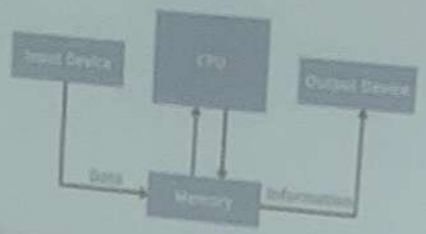
Operation: Processing

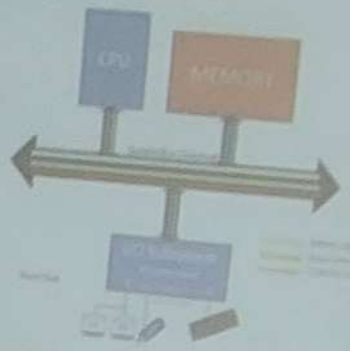
- A computer must be able to process data.
- Data may take a wide variety of forms, and
- The range of processing requirements is broad.
- However, we shall see that there are only a few fundamental methods or types of data processing.



Data Collection

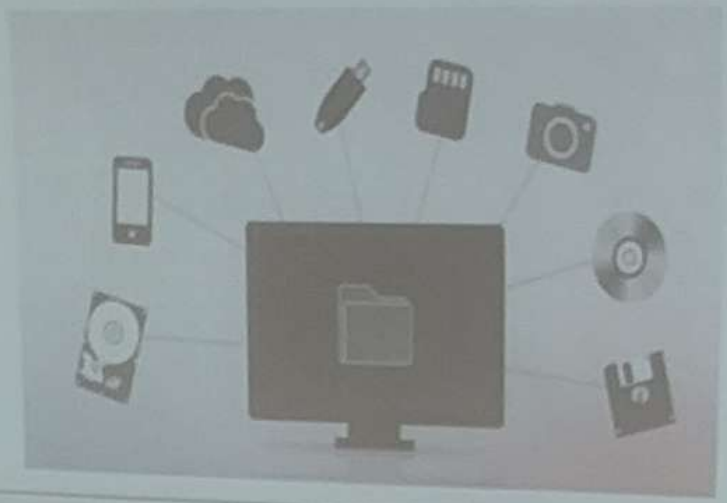
- *Data collection is the process of collecting and evaluating data from multiple sources to accomplish a task.*
- There are two types of devices that allow interaction between the computer and an end user:
  - *input devices and*
  - *output devices.*
- These devices are hardware and component parts of the computer system.
- Both types deal with information and ensure the system works effectively

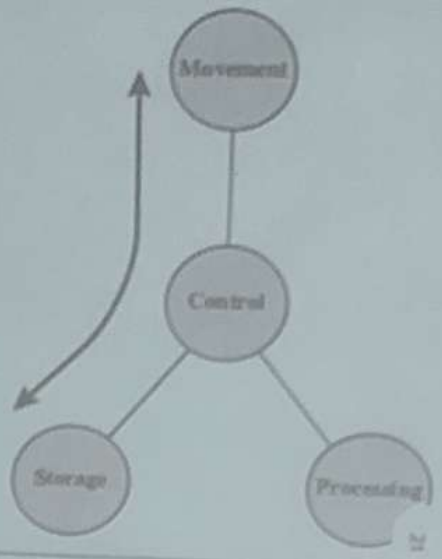




Operation: Storage

- It is also essential that a computer store data.
- Even if the computer is processing data on the fly (i.e., data come in and get processed, and the results go out immediately), the computer must temporarily store at least those pieces of data that are being worked on at any given moment.
- *Thus, there is at least a short-term data storage function.*
- *Equally important, the computer performs a long-term data storage function.*
- Files of data are stored on the computer for subsequent retrieval and update.





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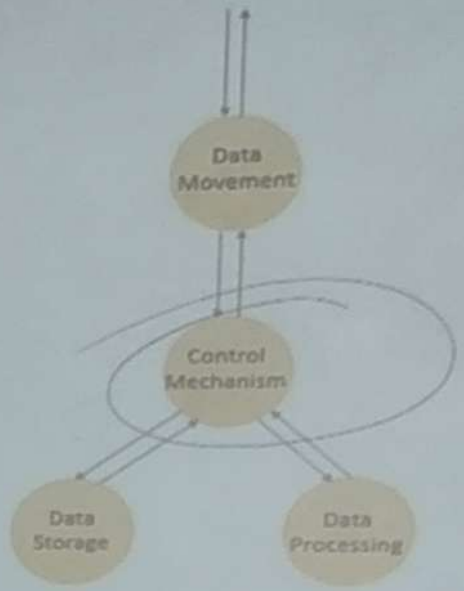
Operation: Control

- There must be control of these three functions.
- Ultimately, this control is exercised by the individual(s) who provides the computer with instructions.
- Within the computer, a control unit manages the computer's resources.
- It orchestrates the performance of its functional parts in response to instructions.